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NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (AIML/CCE) (Credit System) Degree Examinations**

May - June 2023

20AM623 / 20CC623 - AUGMENTED AND VIRTUAL REALITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I**Marks BT* CO* PO***

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|-------|--|---|-----|---|---|
| 1. a) | List modern VR experiences. Explain any six of them. | 8 | L*2 | 1 | 1 |
| b) | Write short note on the following with respect to virtual reality. | | | | |
| | i) First vs. third person | | | | |
| | ii) Synthetic vs. captured world | | | | |
| | iii) Health and safety | 6 | L2 | 1 | 1 |
| c) | Explain the three conditions to be satisfied by the rotational matrix for 2-D rotation. Derive the relation between 2-D rotational matrix and circle with unit radius centered at the origin. | 6 | L3 | 1 | 2 |
| 2. a) | Explain how the idea of 2-dimensional rotation is extended for 3-dimensional rotation by obtaining matrices for roll, pitch and yaw. | 6 | L3 | 2 | 2 |
| b) | Write a short note on chaining of transformations. Discuss transformations applied for left and right eyes. | 5 | L3 | 2 | 2 |
| c) | List and explain the components of look-at. Show the relation between Eye transformation T_{eye} and these components. | 5 | L2 | 2 | 1 |
| d) | Explain canonical view transform. | 4 | L2 | 2 | 1 |
| 3. a) | Explain the following terminologies. | | | | |
| | i) Foveated rendering, | | | | |
| | ii) Display scanout, | | | | |
| | iii) Vergence accommodation mismatch with examples. | 8 | L2 | 2 | 1 |
| b) | Define quaternion. Show the relation between quaternion and axis-angle rotation. Convert the following to axis angle representations / quaternion (0, 0, 0, 1), $(1/\sqrt{2}, 0, 1/\sqrt{2}, 0)$, $((1, 0, 0), \pi)$, $((1, 0, 0), \pi/2)$. | 6 | L3 | 2 | 2 |
| c) | Explain different type of eye movements with examples. | 6 | L2 | 2 | 1 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Discuss different shading methods used for visual rendering with respect to image order rendering. | 8 | L2 | 3 | 1 |
| b) | Discuss the following with respect to rasterization. | | | | |
| | i) Aliasing | | | | |
| | ii) Culling | 6 | L2 | 3 | 1 |
| c) | Explain the following in connection with visual rendering. | | | | |
| | i) Simplifying the virtual world | | | | |
| | ii) Post-rendering image warp | 6 | L2 | 3 | 1 |
| 5. a) | List and explain the four general problems that must be solved to make an effective tracking system with respect to 2D orientation. | 6 | L2 | 3 | 1 |
| b) | List and explain different types of vection. | 6 | L2 | 3 | 1 |
| c) | Explain the Detection mechanism with respect to perception of motion. | 4 | L2 | 3 | 1 |
| d) | Explain the Painter's algorithm and its limitation. How can we overcome the limitation with the help of depth buffer? | 4 | L2 | 3 | 1 |

P.T.O.

6. a) Write a note on the following with respect to tracking 3D orientation.
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|--|---|----|---|---|
| i) Integration | 8 | L2 | 4 | 1 |
| ii) Tilt correction and Yaw correction | 6 | L2 | 4 | 1 |
- b) Write a short note on tracking 2D orientation.
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|--|---|----|---|---|
| | 6 | L2 | 4 | 1 |
|--|---|----|---|---|
- c) List and explain the factors that affect sensitivity.
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|--|---|----|---|---|
| | 6 | L2 | 4 | 1 |
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Unit – III

7. a) Discuss different marker types used in augmented reality applications.
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| | 7 | L2 | 5 | 1 |
|--|---|----|---|---|
- b) List and explain the applications of augmented reality.
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|--|---|----|---|---|
| | 6 | L2 | 5 | 1 |
|--|---|----|---|---|
- c) Define augmented reality. With block diagram explain the components required for AR.
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|--|---|----|---|---|
| | 4 | L2 | 5 | 1 |
|--|---|----|---|---|
- d) List all the steps involved in marker detection procedure.
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|--|---|----|---|---|
| | 3 | L2 | 5 | 1 |
|--|---|----|---|---|
8. a) Briefly discuss the scenarios where marker based solution is a good choice.
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|--|---|----|---|---|
| | 8 | L2 | 5 | 1 |
|--|---|----|---|---|
- b) Briefly explain each of the following:
- | | | | | |
|-------------------------|---|----|---|---|
| i) Model based tracking | | | | |
| ii) Sensor tracking | | | | |
| iii) Hybrid tracking | 6 | L2 | 5 | 1 |
- c) Explain the types of IR markers. Write down the applications and limitations of infrared markers.
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|--|---|----|---|---|
| | 6 | L2 | 5 | 1 |
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
